Luqi Sun

Baltimore, MD, USA

M.S.E. Student in Electrical and Computer Engineering, Johns Hopkins University

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RESEARCH INTERESTS

- **Speech and Language Processing:** I am interested in modeling acoustic and linguistic information to better understand and process human speech and language. My interests in large language models (LLMs) and large audio-language models (LALMs) focus on analyzing internal representations to understand model behavior and developing post-training methods to improve reasoning capabilities and overall performance.
- **AI for Health:** I am interested in applying AI models, including both small-scale and large-scale models, to the assessment of cognitive impairment and mental health, particularly the detection of Alzheimer's disease and depression. I aim to understand how acoustic and linguistic cues reflect health conditions and develop high-performing and reliable models across diverse real-world settings.

EDUCATION

Johns Hopkins University — Baltimore, MD, USA
Master of Science in Engineering in Electrical and Computer Engineering — Aug 2025 – May 2027 (Expected)

Dalian University of Technology — Dalian, Liaoning, China
Bachelor of Engineering in Digital Media Technology — Sep 2021 – Jun 2025

RESEARCH EXPERIENCE

Research Assistant, Johns Hopkins University — Aug 2025 – Present
SMILE Lab, Center for Language and Speech Processing (CLSP)
Advisor: Prof. Berrak Sisman

- Develop speech and language models for Alzheimer's disease and depression detection. Investigate acoustic and linguistic cues and evaluate model robustness across datasets and recording conditions.
- Investigate neuron-level interventions in large language models (LLMs) to understand and control their behavior, with an emphasis on how internal representations influence model outputs. Contribute to LLM development through model experimentation and evaluation of language understanding and generation capabilities.
- Develop post-training methods for large audio-language models (LALMs) to improve their reasoning capabilities and overall performance in healthcare applications.

Research Assistant, Dalian University of Technology — Feb 2024 – May 2025
Advisor: Prof. Qiufen Xia

- Optimized cold-start performance in serverless computing by designing and proposing a resource scheduling system based on a nested architecture. This work resulted in a first-author paper and a granted patent.

PUBLICATIONS

[1] Xiutian Zhao, **Luqi Sun**, Björn Schuller, and Berrak Sisman. Concentrated Causal Influence, Distributed Diagnostic Information: Cognitive Impairment Decisions in Large Audio-Language Models. *Submitted to ARR Rolling Review (Prefered Venue: NAACL 2027)* [Under Review]

[2] **Luqi Sun**, Shreeram Suresh Chandra, Aurosweta Mahapatra, Emily Mower Provost, Brian MacWhinney, and Berrak Sisman. Robust Cross-Domain Speech-Based Alzheimer's Disease Detection via Iterative Adversarial Self-Training. *Submitted to ICASSP*, 2027. [Under Review]

[3] **Luqi Sun**, Shreeram Suresh Chandra, Lin Zhang, You-Jin Li, Brian MacWhinney, Yu Tsao, Emily Mower Provost, and Berrak Sisman. Cleaner Speech, Weaker Generalization: Revisiting Pitt-Derived Benchmarks for Alzheimer's Disease Detection. *AACL-IJCNLP Main Conference*, 2026.

[4] Xiutian Zhao, **Luqi Sun**, Björn Schuller, and Berrak Sisman. Emotion Across Speech and Faces: Shared Affective Mechanisms in Multimodal Foundation Models. *IEEE Spoken Language Technology Workshop (SLT)*, 2026.

[5] Hsiang-Chen Yeh, **Luqi Sun**, Aurosweta Mahapatra, Shreeram Suresh Chandra, Emily Mower Provost, and Berrak Sisman. Who is Speaking or Who is Depressed? A Controlled Study of Speaker Leakage in Speech-Based Depression Detection. *Interspeech*, 2026.

[6] **Luqi Sun**, Qiufen Xia, Xiaolong Zhai, Zhen Feng, and Jiankang Ren. The Optimization of Container Cold Start Latency in Serverless Computing by Nested Architecture. *21st International Conference on Mobility, Sensing and Networking (MSN)*, 2025.

PATENT

Serverless Computing Container Cold Start Method and Related Device Based on Nested Architecture. Chinese Patent **CN120821531B**.